

Algebraic Number Theory

Problem sheet 3

In this sheet, if there is no more explanation, then K denotes a number field of degree n and with r real embedding and s pairs of complex embedding.

1. (3+2 points) Let $\mathcal{O}$ be a Dedekind domain with field of fractions K , and $\mathfrak{a}$ a nonzero fractional ideal of K , (i.e. finitely generated $\mathcal{O}$ -submodule of K), let us recall the inverse of $\mathfrak{a}$ is $\mathfrak{a}^{-1} := \{x \in K \mid x\mathfrak{a} \subseteq \mathcal{O}\}$.
 - (a) Prove that $\mathfrak{a}\mathfrak{a}^{-1} = \mathcal{O}$. Deduce that the set of fractional ideals is a free abelian group generated by prime ideals of $\mathcal{O}$.
 - (b) Show that $\mathfrak{a}^{-1} \cong \text{Hom}_{\mathcal{O}}(\mathfrak{a}, \mathcal{O})$.
2. (3 points) Verify the Chinese Remainder Theorem for Dedekind domains: If $A \triangleleft \mathcal{O}$ is an ideal in the Dedekind domain $\mathcal{O}$ with decomposition $A = P_1^{\nu_1} \dots P_t^{\nu_t}$ as a product of prime ideals then we have

$$\mathcal{O}/A \cong \bigoplus_{i=1}^t \mathcal{O}/P_i^{\nu_i}.$$

3. (3 points) Show that a lattice $\Gamma \subset V$ is complete if and only if V/Γ is compact.
4. (1 point) Show that Minkowski's lattice point theorem cannot be improved, by giving an example of a centrally symmetric convex set $X \subseteq V$ such that $\text{vol}(X) = 2^n \text{vol}(\Gamma)$ which does not contain any nonzero point of the lattice Γ .
5. (2 points) (**Minkowski's Theorem on Linear Forms**) Let $L_i(x_1, \dots, x_n) = \sum_{j=1}^n a_{ij}x_j$ ($i = 1, \dots, n$) be real homogeneous linear polynomials such that $\det((a_{ij}))_{ij} \neq 0$ and let $c_1, \dots, c_n$ be positive real numbers with $c_1 \dots c_n > |\det((a_{ij}))_{ij}|$. Verify that there exist not all zero integers $m_1, \dots, m_n \in \mathbb{Z}$ such that $|L_i(m_1, \dots, m_n)| < c_i$ ($i = 1, \dots, n$). (Hint: Use Minkowski's lattice point theorem)
6. (2+2 points) Show that the map

$$\begin{aligned} \mathbb{C} \otimes_{\mathbb{Q}} K &\rightarrow K_{\mathbb{C}} \\ z \otimes \alpha &\mapsto z \cdot (j\alpha) \end{aligned}$$

is an isomorphism of rings. Moreover, prove that its restriction to $\mathbb{R} \otimes_{\mathbb{Q}} K$ also induces an isomorphism between $K \otimes_{\mathbb{Q}} \mathbb{R}$ and $K_{\mathbb{R}}$.

7. (a) (4 points) Let $X_t := \{z \in K_{\mathbb{R}} \mid \sum_{\tau} |z_{\tau}| < t\}$ for any positive real number t . Verify $\text{vol}(X_t) = 2^r \pi^s \frac{t^n}{n!}$.
- (b) (3+1 points) For any integral ideal $\mathfrak{a}$ (or, more generally fractional ideal). Show that there exists a nonzero $\alpha \in \mathfrak{a}$ such that

$$|N_{K/\mathbb{Q}}(\alpha)| \leq \left(\frac{4}{\pi}\right)^s \frac{n!}{n^n} \sqrt{|d_K|} N(\mathfrak{a})$$

Deduce that $|d_K|^{1/2} \geq \frac{n^n}{n!} \left(\frac{\pi}{4}\right)^{n/2}$. (Hint: choose t so that $\text{vol}(X_t) > 2^n \text{vol}(\Gamma)$ where $\Gamma = j(\mathcal{O}_K) \subset K_{\mathbb{R}}$. Apply the inequality between geometric and arithmetic mean on the numbers $|\tau(\alpha)|$ where $0 \neq \alpha \in \mathcal{O}_K$ is the element with $j(\alpha) \in X_t$ guaranteed by Minkowski's lattice point theorem. Finally, note that $1 \leq |N_{K/\mathbb{Q}}(\alpha)|$.)

- (c) (1 point) Show that whenever the degree $[K : \mathbb{Q}]$ goes to infinity, so does the discriminant d_K . Moreover, we have $d_K > 1$ for all extensions $K \neq \mathbb{Q}$.

Remark. i. This implies that (you will see) if K is a number field such that all prime number p is unramified, then $K = \mathbb{Q}$.

ii. One can prove that for any integer d , there are only finitely many number fields with discriminant d . (We will prove this in next sheet)

(d) (1+1+1 points)(**Minkowski bound**) Prove that every ideal class of K contains an integral ideal $\mathfrak{a}$ with norm:

$$N(\mathfrak{a}) \leq \left(\frac{4}{\pi}\right)^s \frac{n!}{n^n} \sqrt{|d_K|}.$$

Further, prove that the number of integral ideals of $\mathcal{O}_K$ with a given norm is finite. Deduce that its ideal class group Cl_K is a finite abelian group.

8. (1 point for each) Let $K = \mathbb{Q}[\sqrt{d}]$ be an **imaginary** quadratic field, where $d < 0$ is squarefree. Show that $\mathcal{O}_K$ is a PID if d is one of the following integers:

$$-1, -2, -3, -7, -11, -19, -43, -67, -163.$$

Hint: Use Minkowski bound.

Remark. Historically, Gauss proved the above result and conjectured that there was no other such d , which was finally proved by Baker, Heegner and Stark in the middle of the 20th century, after people has worked on it for more than 150 years.

9. (1 point for each) Let $K = \mathbb{Q}[\sqrt{d}]$ be a **real** quadratic field, where $d > 0$ is squarefree. Show that $\mathcal{O}_K$ is a PID if d is one of the following integers:

$$2, 3, 5, 6, 7, 11, 13, 14, 17, 19, 21, 22, 23, 29, 31, 33, 37, 38, 41, 43, \\ 46, 47, 53, 57, 59, 61, 62, 67, 69, 71, 73, 77, 83, 86, 89, 93, 94, 97$$

Hint: Use Minkowski bound.

Remark. It is conjectured that there are infinitely many real quadratic fields of class number 1, which is still an open problem.

10. (a) (3 points) Let $A \triangleleft \mathcal{O}_K$ be an ideal whose class has order m in the class group (ie. m is the smallest integer such that $A^m = (\alpha)$ is a principal ideal). Prove that $A\mathcal{O}_L = (\beta)$ where $L = K(\beta)$ and $\beta^m = \alpha$.
 (b) (1 point) Verify that for each number field K there is a finite extension L/K in which all ideals of $\mathcal{O}_K$ become principal (ie. all ideals of $\mathcal{O}_K$ capitulate in $\mathcal{O}_L$).
 (c) (2 points) Verify that the ring Ω of all algebraic integers (in $\mathbb{C}$) is a *Bézout domain*, ie. all finitely generated ideals are principal (but not noetherian, so not a PID).

11. (a) (3 points) Show that over a Bézout domain every finitely generated torsion-free module is free.

(b) (2 points) Verify that $\bigcup_{n=1}^{\infty} \mathbb{C}[[x^{1/n}]]$ is a Bézout domain.

12. In this exercise we compute the rank of the unit group $\mathcal{O}_K^\times$ as an abelian group using a multiplicative version of Minkowski's theory.

- (a) (3 points) Let $K_{\mathbb{C}}^\times = \prod_{\tau} \mathbb{C}^\times$ be the multiplicative group of invertible elements in the ring $K_{\mathbb{C}}$ and let $N: K_{\mathbb{C}}^\times \rightarrow \mathbb{C}^\times$ be the group homomorphism defined as the product of coordinates. Further define the homomorphism $l := \log |\cdot|: K_{\mathbb{C}}^\times \rightarrow \prod_{\tau} \mathbb{R}$ coordinatewise. Show that the kernel of $l \circ j: \mathcal{O}_K^\times \rightarrow \prod_{\tau} \mathbb{R}$ is the torsion subgroup in $\mathcal{O}_K^\times$, ie. the group $\mu(K)$ of roots of unity in K .
 (b) (2 points) Verify that for each $\alpha \in \mathcal{O}_K^\times$ the element $l \circ j(\alpha) \in \prod_{\tau} \mathbb{R}$ is fixed by the complex conjugation (permuting $\tau \in \text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$). Further show that the sum of coordinates of $l \circ j(\alpha)$ equals 0.
 (c) (5 points) Show that $l \circ j(\mathcal{O}_K^\times)$ is a complete lattice in the subspace H where

$$H = \{x_{\tau} \in \prod_{\tau} \mathbb{R} \mid \sum_{\tau} x_{\tau} = 0 \text{ and } x_{\sigma_k} = x_{\overline{\sigma_k}} \ (k = 1, \dots, s)\}.$$

In particular, $\mathcal{O}_K^\times \cong \mu(K) \times \mathbb{Z}^{r+s-1}$ (as an abelian group).